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James Evans

jevans@uchicago.edu

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machine would, amongst other tables, compute one from the expression

$$A \sin \theta + B \sin 2\theta + C \sin 3\theta$$

the utility of which, in astronomy, is well known. In fact Mr Babbage is of opinion that it would not be impossible to form a machine which should tabulate almost any individual equation of differences.

Amongst the singular and curious powers produced by small additions to the machinery, may be reckoned the possibility of tabulating series expressed by the following equation:

$\Delta^2 u_z$ = the unit's figure of u_z
$\Delta^3 u_z$ = 2× the figures found in the ten's place of u_{z+1}
$\Delta^3 u_z$ = 4× the figures found in the unit's and ten's place of u_{z+1}

and many others similar thereto. /

Again, let the machine be in the act of tabulating any series, a part may be attached by means of which, whenever any particular figure (a 6 for example) occurs in the unit's place, any other number (23 for instance) shall be added to that and all the succeeding terms: and when, in consequence of this, another figure 6 occurs in the unit's place, then 23 more will be added to that and all the succeeding terms. Or, if it be preferred, the number added shall be added to the term ending in 6 only, and not to all succeeding ones.

These views may appear to some persons more curious than useful. They lead however to speculations of a very singular and difficult nature in determining the laws which such series follow: and they are not altogether so remote from utility as may be imagined. I avoid alluding to many other curious properties which this machine is capable of exhibiting, as they will scarcely be intelligible till the machine itself is more known in the world. Indeed I fear I have already tired your patience with this long letter.

London,
28 November, 1823 /

VII

ADDRESS ON PRESENTING THE GOLD MEDAL OF THE ASTRONOMICAL SOCIETY TO CHARLES BABBAGE[a]

BY HENRY THOMAS COLEBROOKE

From *Memoirs of the Astronomical Society*, Vol. 1 (1825), pp. 509–12

This country and the present age have been pre-eminently distinguished for ingenuity in the contrivance, or in the improvement, of machinery. In none has that been more singularly evinced, than in the instance to which I have the gratification of now calling the attention of the Society. The invention is as novel, as the ingenuity manifested by it is extraordinary.

In other cases, mechanical devices have substituted machines for simpler tools or for bodily labour. The artist has been furnished with commands of power beyond human strength, joined with precision surpassing any ordinary attainment of dexterity. He is enabled to perform singly the work of a multitude, with the accuracy of a select few, by mechanism which takes the place of manual labour or assists its efforts. But the invention, to which I am adverting, comes in place of mental exertion: it substitutes mechanical performance for an intellectual process: and that performance is effected with celerity and exactness unattainable in ordinary methods, even by incessant practice and undiverted attention.

[a] H. P. Babbage states in *Babbage's calculating engines* (London, 1889, p. 223) that the medal was awarded 13 July, 1823. This date has been accepted by many authors. The medal itself, however, bears the date 1824, and Colebrooke did not become President of the Astronomical Society until February, 1824. An MS. note on Colebrooke's *Address* in the Babbage Papers, Waseda University, states: '13 February, 1824. The Council awarded their gold medal to Charles Babbage'.

The invention is in scope, as in execution, unlike anything before accomplished to assist operose computations. I pass by, as what is obviously quite different, the Shwanpan, or Chinese abacus, the tangible arithmetic of Frend, Napier's rods, with the ruder devices of antiquity, the tallies, the cheque, / and the counters. They are unconnected with it in purpose, as in form. Mechanical aid of calculation has in truth been before proposed by very eminent persons. Pascal invented a very complicated instrument for the simplest arithmetical processes, addition and subtraction, and reaching by very tedious repetition to multiplication and division. Leibnitz proposed another, of which the power extends no further. Delepine's and Boitissendeau's contrivances, which a century ago were applauded by the Academy of Sciences at Paris, are upon the model of Pascal's, and may no doubt be improvements of it, but do not vary or enlarge its objects. Moreland's instruments, described in an early volume of the *Philosophical Transactions* (the 8th), are confined, the one to addition and subtraction, the other to multiplication. The *Rotula Arithmetica* of Brown, simpler in construction, reaches not beyond the four arithmetical operations.

The principle, which essentially distinguishes Mr Babbage's invention from all these, is, that it proposes to calculate a series of numbers following any law by the aid of *differences*; and that, by setting a few figures at the outset, a long series of numbers is readily produced by a mechanical operation. The method of *differences*, in a very wide sense, is the mathematical principle of the contrivance. A machine to add a number of arbitrary figures together is no economy of time or trouble; since each individual figure must be placed in the machine. But it is otherwise when those figures follow some law. The insertion of a few at first determines the magnitude of the next; and these of the succeeding. It is this constant repetition of similar operations, which renders the computation of tables a fit subject for the application of machinery. Mr Babbage's invention puts an engine in the place of the computer. The question is set to the instrument; or the instrument is set to the question; and, by simply giving it motion, the solution is wrought and a string of answers is exhibited.

Nor is this all; for the machine may be rendered capable of recording its answer, and even multiplying copies of it. The usefulness of the instrument is thus more than doubled: for it not only saves time and trouble in transcribing results into tabular form, and setting types for the printing of the table constructed with them, but it

likewise accomplishes the yet more important object of ensuring accuracy, obviating numerous sources of error through the careless hands of transcribers and compositors. /

On this part of the invention, which is yet a subject of experiment for the selection of the most eligible amongst divers modes of accomplishing it, I shall not dwell longer; as it is not for that superaddition, but for the machine in the finished form of a calculating instrument, that I am to make an acknowledgement, in the name of this Society, to Mr Babbage for his very useful invention.

I speak of it as complete with reference to a model which satisfactorily exhibited the machine's performance, and am apprised that a more finished engine, which is in progress of preparation, may not yet for some time be in a forward state to be put in activity and receive its practical application.

In no department of science or of the arts, does this discovery promise to be so eminently useful as in that of astronomy and its kindred sciences, with the various arts dependent on them. In none are computations more operose than those which astronomy in particular requires: in none are preparatory facilities more needful: in none is error more detrimental. The practical astronomer is interrupted in his pursuit, is diverted from his task of observation, by the irksome labour of computation; or his diligence in observing becomes ineffectual for want of yet greater industry of calculation. Let the aid, which tables previously computed afford, be furnished to the utmost extent which mechanism has made attainable through Mr Babbage's invention, the most irksome portion of the astronomer's task is alleviated, and a fresh impulse is given to astronomical research.

Nor is it amongst the least curious results of the ingenious device, of which I am speaking, that it affords a new opening for discovery; since it is applicable, as has been shown by its inventor, to surmount novel difficulties of analysis. Not confined to *constant differences*, it is available in every case of differences that follow a definite law, reducible therefore to an equation. An engine, adjusted to the purpose, being set to work, will produce any distant term, or succession of terms, required: thus presenting the numerical solution of a problem, even though the analytical solution of it be yet undetermined.

It may not therefore be deemed too sanguine an anticipation, when I express the hope, that an instrument, which in its simpler form attains to the / extraction of the roots of numbers and approximates

to the roots of equations, may in a more advanced state of improvement rise to the approxiate solution of algebraic equations of elevated degrees. I refer to solutions of such equations proposed by Lagrange, and more recently by other analysts, which involve operations too tedious and intricate for use, and which must remain without efficacy, unless some mode be devised of abridging the labour or facilitating the means of its performance. In any case this engine tends to lighten the excessive and accumulating burden of arithmetical application of mathematical formulae, and to relieve the progress of science from what is justly termed by the author of this invention, the overwhelming incumbrance of numerical detail.

For this singular and pregnant discovery, I have the authority of the Astronomical Society of London to present to Mr Babbage, its Gold medal, which accordingly I now do, as a token of the high estimation in which it holds his invention of an engine for calculating mathematical and astronomical tables. /

VIII

ON THE DETERMINATION OF THE GENERAL TERM OF A NEW CLASS OF INFINITE SERIES

From *Transactions of the Cambridge Philosophical Society*, Vol. 2 (1826), pp. 217–25

Read 3 May, 1824[a]

The subject of investigation on which I have entered in the following paper, had its origin in a circumstance which is, I believe, as yet singular in the history of mathematical science, although there exists considerable probability, that it will not long remain an isolated example of analytical enquiries, suggested and rendered necessary by the progress of machinery adapted to numerical computation. Some time has elapsed since I was examining a small machine I had constructed, by which a table, having its second difference constant, might be computed by mechanical means. In considering the various changes which might be made in the arrangement of its parts, I observed an alteration, by which the calculated series would always have its second difference equal to the unit's figure of the last computed term of the series; other forms of the machine would make the first or the third, or generally any given difference equal to the unit's figure of the term last computed; and a further alteration would make the same difference equal to double, or generally to (a) times the digit in the unit's place: or if it were preferred, / the digit fixed upon might be that occurring in the ten's place, or generally in the nth place. I did not, at that time, possess the means of making these

[a] Although this paper was read 3 May, 1824, the date of publication is uncertain. The date 1826, used here, is suggested by A. W. Van Sinderen in 'The printed papers of Charles Babbage', *Annals of the History of Computing*, Vol. 2. (1980), pp. 169–85.